

Benjamin Wang

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EDUCATION

University of Virginia

Bachelor of Science in Computer Science

Charlottesville, VA

May 2025

EXPERIENCE

MacChat

June 2025 – Present

Rotational Engineer, PennyMac

Cary, NC

- Developed and maintained the evaluation suite for PennyMac's Servicing Chatbot, based in Python/PostgreSQL
- Engineered new pipelines to track MacChat metrics such as response times, load capabilities, and performance volatility, leading to a reduction of model-degradation debugging time from 1-2 weeks to 1-3 days
- Consolidated numerous isolated accuracy tests/ dataset-specific pipelines into a single, modular framework that supports plug-and-play datasets, reducing the time to onboard a new dataset from 2-3 days to less than 1
- Authored technical documentation and gap analysis, speeding onboarding within the development team by 50% and knowledge transfer times across company teams by 95%

Audio-Visual Technical Assistant

September 2022 – May 2025

University of Virginia

Charlottesville, VA

- Delivered technology/troubleshooting support for audio-visual equipment for campus university events
- Enhanced IT and problem solving skills by resolving real-time tech issues under dynamic/high pressure situations
- Cultivated strong interpersonal and conflict-resolution skills through interactions with a wide range of stakeholders, spanning student clubs, university leadership, and even state governance

PROJECTS

ML4VA: Predicting UVA Course Difficulties | *Python, scikit-learn*

October 2024 - December 2024

- Designed a Random Forest model to help predict course GPA's at UVA, achieving an overall RMSE of ± 0.2
- Preprocessed a dataset of 29,964 entries, using techniques such as imputation, standard scaling, and one-hot encoding on categorical data to optimize model performance
- Planned and produced a comprehensive presentation video showcasing the project's process, findings, and insights, winning second place in the ML4VA Fall 2024 Competition

Portfolio Website | *Javascript, React, TailwindCSS, Next.js*

July 2024

- Developed a single page, modal-like portfolio from scratch using Next.js and React
- Explored visual design and layout concepts to craft a portfolio that is both creative and aesthetically pleasing
- Designed and built (mostly) all UI components from scratch, implementing custom layouts, modals, and animations to achieve a very personal portfolio

Wordle Screenshot Parser | *Python, OpenCV, TesseractOCR, NiceGUI*

May 2024 - August 2024

- Designed and implemented a Python-based data analytics tool responsible for automatically extracting/parsing game information from Wordle screenshots using TesseractOCR
- Integrated a series of smart filters and correction systems to handle several types of edge cases, increasing screenshot detection rate to 96% and accuracy rates to 85%

SKILLS

Languages: Python, SQL, Java, JavaScript, C

Web Development: HTML/CSS, Django, Express.js, Next.js, React, TailwindCSS

Tools and Libraries: NumPy, OpenCV, TesseractOCR, JUnit, FXML, Pygame

Softwares/Platforms: Cursor, Microsoft Office, Google Office Suite, Adobe Photoshop and Premiere, Godot